VISWANADHA ANURAG VISAGAKOTI

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SUMMARY:
Robotics & Autonomous Systems Engineer with 12+ years of experience spanning robot manipulation, aerial autonomy, motion planning, and Embodied AI. Combines rigorous foundations in multi-body dynamics, inverse kinematics, visual servoing, and ROS/ROS2 with modern 3D spatial perception (Gaussian Splatting, NeRFs, SLAM) and deep learning model optimization & edge deployment. Author of peer-reviewed robotics research at IEEE ICRA and IROS. Proven track record taking autonomous systems from mathematical formulation and Isaac Sim/Gazebo simulation to field-deployed physical hardware (PX4 UAVs, industrial manipulators, and edge microcontrollers).

SKILLS

- **Robotics Areas**: SLAM, Kinematics & Multi-Body Dynamics, Inverse Kinematics (IK), Motion Planning, Visual Servoing, Sensor Fusion, Manipulator Control.
- **Robotics Tools**: ROS/ROS2, Nav2 (Navigation Stack), MoveIt, Flight Control (PX4 / MAVROS), NVIDIA Isaac Sim, Gazebo, RViz, MATLAB / Simulink, MuJoCo, SolidWorks(CAD).
- **Sensors & Hardware**: LiDAR, RGB-D / Stereo Cameras, Vicon MoCap Systems, IMU, ESP32 Microcontrollers, PLC / HMI
- **3D Perception & Computer Vision**: SLAM (ORB-SLAM3, GS-SLAM, LiDAR SLAM), 3D Gaussian Splatting, NeRF, TorchVision, Albumentations, PiFUHD, YOLO (YOLOv8-seg), U-Net, PointNet++, RT-DETR, OpenCV, Point Clouds, Line 3D++, g2o, GTSAM, Ceres Solver, PCL.
- **MLOps & Edge Deployment**: TensorRT (FP16 / Engine Compilation), ONNX Runtime, CUDA Kernel Optimization, Dynamic Gaussian Pruning, Docker, FastAPI, AWS (EC2 G4dn/G5), Google AI Edge LiteRT, TinyML.
- **Agentic AI, Orchestration & Retrieval**: LangGraph (StateGraph), LangChain (LCEL), Vercel AI SDK, ChromaDB, Pydantic v2, Tenacity (Resilience), FastAPI, LiteLLM, Ollama.
- **Languages & Systems**: C++, Python, MATLAB, CUDA, SQL, Linux / Shell, Docker, FastRTPS / WebSockets, Git, AWS (EC2 G4/G5).

PUBLICATIONS

- [1] Rachit Bhargava, P.Mithun, V. V. Anurag, A.H.Abdul Hafez, S. V. Shah "Image Space based Path Planning for Reactionless Manipulation of Redundant SpaceRobot" IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS),Korea, October 2016.
- [2].Shah, S. V., Viswanadha Visagakoti, Anurag, Abdul Hafez, A. H., "Switching method to Avoid Algorithmic Singularity in Vision-based Control of a Space Robot", International Conference on Advanced Robotics(ICAR), IEEE, May 2015, Istanbul, Turkey
- [3] Abdul Hafez, A. H., Anurag, V. V., Shah, S. V., Krishna, K. M., Jawahar, C. V."Reactionless visual servoing of a dual-arm space robot", International Conference on Robotics and Automation(ICRA), IEEE,May 2014, Hong Kong, China

INDUSTRY EXPERIENCE

AI enabled Project coordinator: NirvanaHealth | RxAdvance, Southborough, USA Oct 2025 – Current
Agentic AI automation layer for Azure Boards

- Engineered a distributed **multi-agent execution** pipeline with **self-healing feedback loops** and **dynamic error recovery**, achieving 99.9% schema compliance against strict **Pydantic** v2 data contracts.
- Architected a **zero-egress, air-gapped LLM** inference system on local hardware, **optimizing prompt ingestion** pipelines to cut token overhead by 82% and eliminate network latency.

Software Engineer: Technofix IT Solutions, Tempe (Remote), USA July 2025 – Sept 2025
Voice assistant agentic AI

- Engineered an **autonomous conversational agent loop (chatbot)** using **Vercel AI SDK**, executing **dynamic multi-step tool calls** for timers, reminders, and search.
- Implemented a **streaming agent orchestration layer** with dynamic tool execution and boundary clipping to optimize text-to-speech synthesis.
- Built an end-to-end voice pipeline integrating **Whisper/Fish-Audio STT, LLM reasoning, and OpenAI TTS** with real-time PCM16 audio resampling.

Monitoring Analyst: Public Partnerships LLC, Tempe (Remote), USA July 2024 – July 2025

Payroll processing job automation

- Engineered **SQL-based automation pipelines** for end-to-end auditing of timesheets, fund allocation, and payment disbursement, reducing manual intervention by over 60%.
- Designed **Python scripts** for parsing NAV audit logs, generating real-time status reports, and identifying high-priority exceptions using regex and structured log aggregation.

AI/Robotics Research assistant: Arizona State University, Tempe, USA June 2023 – July2024

3D Gaussian Splatting Reconstruction & Differentiable Path planning & optimization.

- **Real-Time 3D Scene Reconstruction:** Optimized training of 3D Gaussian Splatting (3DGS) representations directly from incoming robot camera video streams, enabling incremental scene reconstruction and dynamic spatial mapping during active navigation.
- **Differentiable Trajectory Optimization**: Re-engineered the GPU-accelerated 3DGS tile-based rasterization pipeline to project 3D Gaussian covariance ellipsoids into volumetric density fields, utilizing gradient-based optimization for real-time obstacle avoidance and collision-free path generation.
- **Gradient-Based Path Planning**: Formulated an end-to-end differentiable path planner in PyTorch that iteratively deforms candidate trajectories to minimize path length, kinodynamic curvature, and obstacle collision probabilities.
- **Active Perception & Real-Time Execution**: Implemented active viewpoint planning to navigate the robot toward unmapped regions, accelerating 3DGS scene convergence while enforcing real-time compute budgets for closed-loop obstacle avoidance.
- **Benchmarking & Validation**: Curated synthetic multi-view spatial navigation datasets in PyTorch, validating trajectory smoothness, clearance metrics, and plan optimality against baseline graph search (A*, RRT*) and kinodynamic benchmarks.

Staff Engineer: Treadstone Media Labs, Bengaluru, India March 2022 – May 2022

3D Model generation for graphics tool research and development.

- Led **end-to-end planning and execution** of **interactive graphics tooling**, coordinating research, development, and deployment milestones for a prototype-grade animation pipeline used by creative and procedural content teams.
- Engineered GPU-accelerated 3D asset processing pipeline (mesh denoising, watertight reconstruction, adaptive remeshing), significantly reducing downstream cleanup time for artists.
- Planned and delivered **deep learning–based 3D asset generation tools** using **PiFUHD** for 3D digital human models from 2D images.
- **Optimized** the PiFUHD deep implicit surface network by converting **PyTorch models to ONNX** and compiling TensorRT FP16 inference engines, cutting per-image 3D reconstruction latency.
- Defined **data strategy and execution plan** for a proprietary 3D dataset (>100K meshes), coordinating **dataset curation, augmentation, and model fine-tuning** (**transfer learning, LoRA**) to meet quality and generalization targets.
- Architected and deployed a containerized GPU inference service on AWS EC2 (G4dn/G5 instances) using Docker and FastAPI with asynchronous worker queues.

Lead Engineer: Orange Bodhi Technologies, Hyderabad, India Apr 2020 – March 2022

Development of Industrial robotics applications

- Architected and deployed a fully automated **multi-stage chemical etching system** using a **4-DOF robotic manipulator** to transport metal substrates through sequential acid bath treatments, each requiring dip time and agitation profiles.
- Developed a **state-machine based process controller** that synchronized robotic arm motions with tank-specific safety interlocks and dynamic timeout triggers based on system sensor feedback.
- Engineered **stereo vision-based surface inspection** modules using calibrated industrial cameras and custom OpenCV pipelines to detect surface inconsistencies (bubbles, discoloration, erosion), enabling closed-loop feedback to optimize dip durations and motion paths.
- **Fine-tuned** a deep segmentation model (**U-Net / YOLO-seg backbone**) in PyTorch on a custom 15k-image industrial dataset with extensive photometric and geometric augmentations (Albumentations), achieving 98.2% mAP for micro-defect and bubble detection.
- Integrated the vision inference pipeline with **ROS-based motion planning** to enable closed-loop visual feedback, dynamically adjusting robotic arm dip durations and trajectory offsets based on real-time surface defect segmentation.
- Implemented **dynamic robot path planning** routines using **inverse kinematics in ROS** to adapt to conveyor line offsets, ensuring consistent submersion across variable workpiece geometries.

System Engineer: **TCS Innovation Labs, Kolkata, India** March 2017 – April 2020

Retail Space Automation with quadrotors

- Conducted R&D of **3D perception** and **navigation** for quadrotor-based retail space automation.
- Designed and constructed a **high-fidelity indoor testing facility** equipped with a **Vicon motion capture system** for ground-truth trajectory recording, enabling quantitative evaluation of localization accuracy, loop closure, and drift correction in real-time.
- Tested SOTA **path planning(A*)** and **trajectory control** to operate indoor quadrotors using motion capture feedback.
- Modified **ORB-SLAM** code for higher feature output for line-based object segmentation pipelines (**Line 3D++**).
- Integrated **DfUSMC (Depth-filtered Unsupervised Surface Mesh Completion)** into the navigation stack, enabling detailed 3D reconstruction during hover states and fly-throughs, supporting semantic segmentation.
- Developed a telemetry bridge between **PX4 autopilot firmware** and the perception stack using **MAVROS**, allowing direct interfacing of high-level motion commands from SLAM to the flight controller.

Robotics Engineer: **Asimov Robotics, Kochi, India** Feb 2016 – Feb 2017

Design and development of PR humanoid robot with custom manipulators

- Led the **end-to-end mechanical design and control integration** of a humanoid robot targeted for public interaction roles in malls, hospitals, and expos, focusing on aesthetic ergonomics and dynamic functionality.
- Implemented **manipulator control** for basic tasks considering payload by formulating manipulator dynamics.
- Executed **2D LiDAR SLAM** based navigation based on pre-built map to provide customer assistance.

Research Assistant: **IIIT Hyderabad, Hyderabad, India** Feb 2012 – Jan 2016

Floating base Dual arm space manipulator-Kino dynamic Simulation, planning and control

- Research involved **Task priority-based manipulation**, **multi-body dynamics** formulation, **visual servo control**, **Planning** in image space for a floating base redundant dual-arm space manipulator. ICRA[3], IROS[1], ICAR[2].
- Innovated new methods in calculation of **dynamic** decoupling of floating base disturbance and dual arm manipulator.
- Incorporated in MATLAB simulation using numerical methods and employed **visual servo control**
- Employed image space path planning for optimized execution while minimizing base disturbance.
- Constructed a prototype to demonstrate gravity free conditions on a hyper-smooth granite slab using air pressure knobs.

Product Design Engineer: **Purnansh Design, Kochi, India** Jun 2010 –Dec 2011

Product Design: Aesthetic, functional and CAD Design

- Designed consumer-grade and industrial products with a focus on manufacturability, integrating both aesthetic form and functional mechanical performance.
- Created detailed 3D CAD models and production-ready drawings in **SolidWorks**, applying GD&T and tolerance stack-up analysis to ensure part interchangeability.
- Collaborated with fabrication vendors to translate digital models into prototypes using injection molding, sheet metal, and 3D printing workflows.
- Conducted design validation via **FEA simulations**, ergonomic analysis, and DFM/DFA reviews to balance structural integrity with visual appeal.

EDUCATION

- **M.S. Robotics and Autonomous Systems (Systems Engineering)** May 2024
- Arizona State University, Tempe, AZ
- **M.S. Computer Science and Engineering** July 2015
- International Institute of Information Technology (IIIT), Hyderabad, India
- **B.Tech. Mechanical Engineering** June 2010
- Vellore Institute of Technology (VIT), Vellore, India